Algorithmic Mountain Bike (MTB) Trail Difficulty Classification

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***Abstract*—Objectively rating mountain bike (MTB) trail difficulty remains an open problem, as existing systems rely on subjective, locally applied criteria that vary across regions. This paper investigates automated difficulty classification using features derived from GPX files recorded by consumer GPS devices. Three models are compared on a dataset of 34,869 community-rated trails consolidated into four difficulty classes: a multi-layer perceptron baseline, a convolutional neural network trained on rasterized trail images, and a Bayesian classifier with interpretable posterior weight distributions. All three converge on test accuracies of 0.44–0.50, exceeding the 0.25 random baseline but suggesting a performance ceiling imposed by the partial observability of difficulty-relevant trail characteristics in GPS data. The Bayesian model's comparable accuracy using only four gradient and elevation features indicates these statistics capture the bulk of the difficulty signal available in GPX recordings.**

***Index Terms*—mountain biking, trail difficulty classification, GPX data, GPS trajectory analysis, convolutional neural network, Bayesian classification, multi-layer perceptron, ordinal classification**

I. Introduction

TRAIL rating remains a difficult and inherently human process. Across a variety of outdoor activities (e.g. hiking, cycling, and skiing), difficulty ratings rely on relative systems defined and applied locally, preventing trail users from comparing difficulties between areas. At worst, this inconsistency poses a safety concern: a rider or hiker who performs competently at a given rating in one region, may encounter dramatically more challenging terrain under the same rating elsewhere, leading to potential injury. Adopting a more algorithmic approach, driven by quantifiable trail features and standardized across regions, would allow users to access appropriate difficulty information and make informed decisions that support both safety and activity enjoyment.

The space of categorizing mountain biking (MTB) trails specifically is of interest. The International Mountain Bicycling Association (IMBA) provides a widely referenced rating scale, yet its application is ultimately left to local trail builders and land maintainers, introducing the subjectivity and regional inconsistencies observed in other disciplines. Further, compared to hiking or skiing, relatively little prior work has explored data-driven approaches to MTB trail classification, leaving the problem largely unaddressed in the literature.

A promising data source for this task is the GPX file format. GPX files are routinely recorded by riders using consumer GPS devices and encode georeferenced sequences of latitude, longitude, and elevation at high temporal resolution. From these traces, features relevant to perceived difficulty, such as total elevation gain, grade, elevation variability, and descent profiles, can be extracted without requiring specialized survey equipment. The ubiquity of GPX data, combined with platforms that aggregate user-recorded rides alongside community-assigned ratings, creates an opportunity to build labeled datasets at scale.

This work presents a two-pronged modeling approach to automated MTB trail difficulty classification. A convolutional neural network (CNN) is trained to learn spatial patterns directly from processed GPX representations, while a Bayesian model offers a complementary probabilistic framework that quantifies uncertainty in its predictions, especially valuable given the subjectivity inherent in trail ratings. To demonstrate the practical applicability, the resulting models are integrated into a web application that allows a user to upload a GPX file and receive a predicted difficulty rating.

II. Background

A. Trail Difficulty Rating Systems and Their Limitations

The challenge of objectively categorizing MTB trails has produced a range of rating systems worldwide, none of which have achieved universal adoption or consistency. The IMBA Trail Difficulty Rating System, adapted from the National Ski Areas Association's terrain rating scheme, remains the most widely referenced framework in North America [1]. The IMBA system evaluates trails based on several criteria including tread surface, average and maximum gradient, trail width, natural obstacles, and technical trail features [1]. However, the system explicitly focuses on technical challenge alone and excludes physical exertion, trail length, and elevation change from its scope. The IMBA's own guideline recommends that trails be rated relative to other trails in the same region, an instruction that by design introduces geographic inconsistency [1]. This fragmentation exists internationally as well, with Europe using at least fourteen distinct rating systems, which rely on similar criteria [2] [3]. Additionally, Siebert et al. [4] found significant differences in perceived trail difficulty across skill levels, underscoring the safety consequences of inconsistent or subjectively applied ratings.

B. Physical Determinants of Trail Difficulty

Understanding which physical trail characteristics contribute most to perceived difficulty guides selection of appropriate input features in any predictive model. Steyn at al. [5] provided foundational work by applying vehicle-pavement interaction (V-PI) principles to mountain bike trail classification in South Africa, concluding that trail gradient is the single greatest contributor to difficulty in terms of physical exertion, while trail roughness, quantified using the International Roughness Index (IRI), directly affects riding quality and required technical skill [5]. Their work also demonstrated that the LS-factor (a composite of slope length and steepness commonly used in erosion modeling) and landform classes serve as significant predictive variables for trail conditions [5].

For a GPX-based modeling approach, it is important to note many of these features identified in the broader trail difficulty literature, such as trail width, natural obstacles like rocks and roots, technical trail features including drop-offs and elevated structures, and tread surface material, are not directly observable from GPS data [1] [2] [6]. However, features the GPX data can capture, including elevation profiles, gradient distributions, and steepness, correspond to several of the most influential difficulty determinants identified in the literature. This partial observability motivates a modeling approach that maximizes the information extractable from elevation and positional data, while acknowledging the inherent limitations of the data source.

C. GPS Trajectory Data as a Modeling Input

The GPX (GPS Exchange Format) file encodes georeferenced sequences of latitude, longitude and elevation recorded at high temporal resolution by consumer GPS devices. While no prior work has applied GPS races specifically to mountain bike difficulty classification, a substantial body of research has demonstrated the efficacy of GPS trajectory data for related classification tasks, establishing both the value of the data source and effective preprocessing strategies.

Dabiri and Heaslip [7] developed a novel representation for raw GPS trajectories, combining the data with a feature vector of motion-related characteristics and roadway-geometric information to produce a matrix representation suitable for deep learning. Their CNN-based vehicle classification model (CNN-VC) significantly outperformed both classical supervised learning algorithms and other deep learning baselines, which Dabiri et al. [8] further demonstrated, proving these raw GPS trajectories contain sufficient descriptive information for nuanced classification tasks when properly represented.

D. Convolutional Neural Networks for Spatial Data Classification

A common thread across successful spatial classification systems is the transformation of non-image data into structured representations suitable for convolution. Langer et al. [9] performed the first computational classification of MTB downhill trail difficulty, using 3D accelerometer and gyroscope data from wearable sensor units. By stacking these sensor streams into a two-dimensional image-like representation, their 2D CNN achieved a Sparse Categorical Accuracy of 0.9097 across three difficulty classes (easy, medium, hard) [9]. Their results demonstrate that two-dimensional representations are highly effective for capturing the physical characteristics of trail segments. However, a critical limitation of their approach is the reliance on specialized inertial measurement unit (IMU) data, which requires purpose-built hardware and cannot be extracted retroactively using widely available ride data.

In climbing research, Dobles et al. [10] projected climbing hold configurations from the Moonboard (a standardized 18x11 climbing wall) into a binary density image and applied a CNN with an ordinal output layer to classify route difficulty. The CNN demonstrated the best combination of accuracy and faithful reproduction of the underlying grade distribution among the methods tested (which also included Naive Bayes and softmax regression), highlighting the inherent difficulties of classifying subjectively assigned grades, a challenge directly parallel to the MTB rating problem [10]. Subsequent work by Chang et al. [11] improved upon these results by preprocessing Moonboard routes into movement sequences that mimic human climbing patterns, achieving near-human-level grade accuracy prediction with a recurrent neural network, suggesting that domain-informed data representations are key to improving classification in subjectively graded activities.

III. Methodology

A. Dataset Collection and Breakdown

The dataset for this study was sourced from the MTB Project, a community-driven platform on which riders upload GPS recordings of MTB trails alongside user-voted difficulty ratings. MTB Projects serves as a crowd-sourced repository aiming to catalog all available MTB trails, making it well-suited as a labeled data source for supervised classification. Although the platform hosts trails internationally, this study was restricted to trails within the contiguous United States to reduce variability in how difficulty ratings are applied.

MTB Project employs a six point ordinal difficulty scale: Easy, Easy/Intermediate, Intermediate, Intermediate/Difficult, Difficult, and Very Difficult. Each trail's displayed rating is determined by the mode of all user-submitted votes, reflecting the community consensus rather than any single individual's assessment. This labeling approach introduces a degree of noise inherent to subjective rating tasks (II-B) but also provides a form of natural aggregation that mitigates individual rater bias.

The total dataset consists of 34,869 trails with both valid GPX data and at least one user-submitted difficulty rating. Trails without any difficulty rating were excluded. Each GPX file encodes a sequence of track points containing latitude, longitude, and elevation; timestamp data was also present in the files but was not utilized in this study as the points were recorded at approximately consistent time intervals and temporal information was not deemed relevant to the difficulty classification task. The number of GPS track points per trail varied considerably, reflecting the wide range of trail lengths

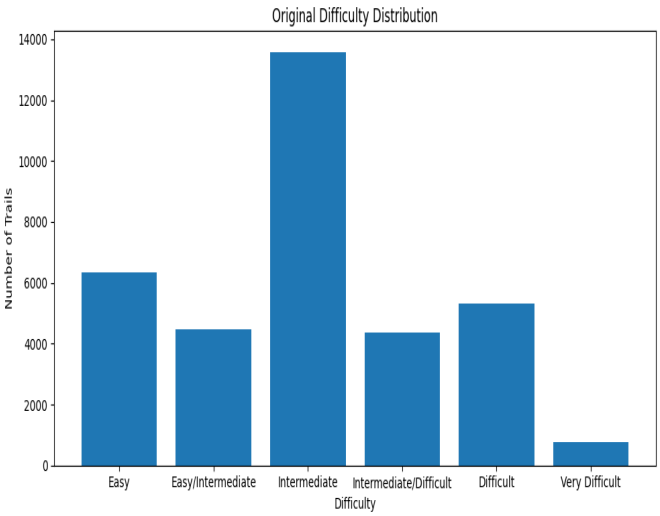

Fig. 1: Original dataset class distributions

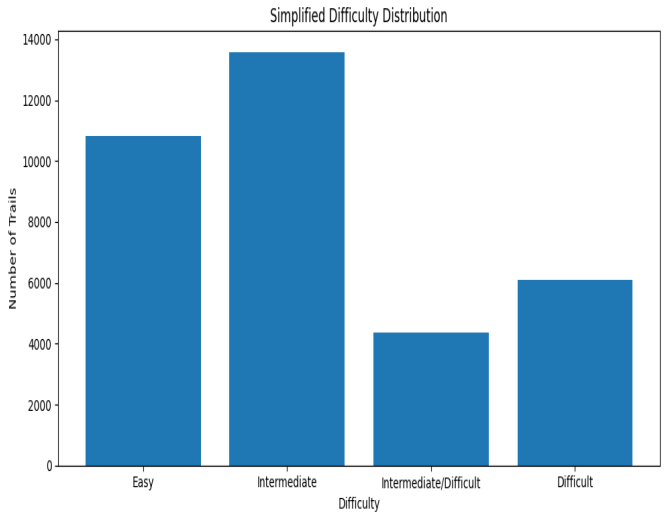

Fig. 2: Dataset class distributions after combining classes

in the dataset (min: 2; max: 3,829; mean: 74.4; median: 44; IQR: 24 - 85). This high variability in sequence length poses a representation challenge for fixed-input models such as CNNs, and is discussed later.

For each trail, a set of general route statistics was computed directly from the raw GPX track points, including total distance, total elevation gain and loss, and average and maximum gradient. These features were calculated programmatically from the GPS track points and were verified against the statistics displayed on MTB Project's trail pages to ensure reasonable accuracy.

The original six difficulty classes exhibited significant class imbalance (1), with Intermediate trails comprising the largest share (about 13,600 trails) and Very Difficult trails the smallest (800 trails). To reduce the number of target classes and mitigate extreme imbalance at the tails, the six-point scale was consolidated into four classes by merging adjacent categories. This consolidation improved balance at the extremes, particularly by absorbing the sparse Very Difficult class into a larger Difficult category, while preserving the ordinal structure of the rating scale (2). However, a roughly 3:1 ratio persisted between the largest class (Intermediate) and the smallest (Intermediate/Difficult). To ensure that no class dominated the training process, the classes were balanced via a random subsampling prior to model training, reducing each class to match the size of the smallest class. This approach sacrifices some training data in the majority classes in exchange for an unbiased class prior during learning, preventing the model from defaulting to majority-class predictions.

B. CNN-Based Classification

To transform variable-length GPS traces into fixed-sized representation suitable for convolutional processing, each trail was rasterized into a 256x256 single-channel grayscale image. The rasterization procedure encodes both the spatial shape of the trail and its elevation profile into a single representation, allowing the CNN to learn joint spatial-elevation patterns through standard convolution operations.

For each trail, the latitude and longitude coordinates of the GPS track points were independently normalized to fill a 208x208 drawable region centered within the 256x256 canvas (the remaining border provides padding to prevent edge artifacts during the convolution). The normalization maps each axis independently to the full drawable range, meaning the trail's geographic aspect ratio is not preserved. While this distortion sacrifices absolute geographic shape fidelity, it ensures that every trail occupies the full spatial resolution of the image regardless of its physical dimensions, maximizing the information available to the convolutional filters.

Pixel intensity encodes relative elevation. Each trail's elevation values are normalized to its own minimum and maximum, mapping the lowest point to 0 (black) and highest point to 255 (white). The background of the image is filled with the trail's mean normalized elevation rather than zero, so that the trail line is drawn atop a canvas whose brightness reflects the trail's average altitude. The trail is rendered by drawing Bresenham line segments between consecutive GPS points using a line width of 4 pixels (producing a 7x7 square brush), with a circular dot of radius 3 pixels (7-pixel diameter) stamped at each GPS location. At self-crossing points, later segments overwrite earlier ones with no blending or transparency, deemed acceptable given that the elevation at a crossing is typically similar in both directions.

While the rasterized image captures spatial and elevation patterns, certain aggregate trail statistics that are known to correlate with difficulty (as discussed in Section II-B) are not easily recoverable from the image alone. Four supplementary numerical features were therefore computed from the raw GPX data for each trail: maximum grade, average grade, total elevation gain, and total elevation loss. These features are supplied to the model as a secondary input and concatenated

with the image-derived features after the convolutional stage, allowing the dense layers to reason over both spatial patterns and global route statistics. These supplementary features were normalized via MinMax Scaling prior to concatenation with the flattened convolutional output.

The CNN architecture has two layers of inputs; the first input layer accepts the 256x256x1 grayscale trail image while the second accepts the four-dimensional supplementary feature vector. The convolution branch consists of two convolutional layers. The first applies 1 filter with a 3x3 kernel, same padding, followed by 5x5 max pooling. The second applies 2 filters with a 3x3 kernel and valid padding, again followed by 5x5 max pooling. Both convolution layers have a stride of 1 and use ReLU activation. The output of the convolution branch is flattened and concatenated with the supplementary feature vector. The combined representation then passes through three fully connected layers with 128, 64, and 32 units respectively. Each dense layer uses Leaky ReLU activation (α = 0.1) to mitigate the dying ReLU problem. Each dense layer is followed by dropout with a rate of 0.7, an aggressive choice aimed at controlling the model's tendency to overfit based on hyperparameter tuning. The output layer consists of a three-unit dense layer with softmax activation, producing a probability distribution over the four consolidated difficulty classes. The model is compiled with categorical cross-entropy loss and the Adam optimizer.

C. Bayesian-Based Classification

The computational cost of Bayesian sampling algorithms motivated a simpler feature set than the CNN. Four trail-level features were used: elevation gain, elevation loss, average grade, and max grade, aligning with the difficulty-relevant variables identified in prior work [5].

Isotropic Gaussian priors (mean = 0, σ = 3) were placed on all weights, reflecting initial uncertainty about feature influence while allowing the data to drive the posterior. The symmetry of the Gaussian prior simplifies the Metropolis-Hastings to a standard Metropolis sampler, as the proposal ratio reduces. A proposal standard deviation of 0.5 was selected, tuned to achieve an acceptance rate between 20% and 30% from the Markov chain. The prior sigma of 3 provides moderate regularization where large weights supported by the data remain unconstrained while small, noisy weights shrink towards zero.

To address strong autocorrelation observed in the chain, a thinning ration of 200 was applied alongside a burn-in to discard pre-convergence samples. These requirements necessitated running the sampler for 1.5 million iterations, of which 6,150 posterior samples were retained. All likelihood and acceptance ratio computations were performed in log space to avoid floating-point overflow.

IV. Results

The MLP model serves as a baseline against which the more complex CNN and Bayesian approaches are evaluated. Using elevation, latitude, and longitude as input features with default hyperparameters, the MLP achieves a training accuracy of 0.534 and a test accuracy of 0.500 over 3,092 (training + validation) input vectors respectively, with a grand total of 21,923 trainable parameters. This makes the MLP the strongest-performing model by test accuracy; it suffers from reliability however as two consecutive runs during validation produced accuracies of 0.500 and 0.250, indicating sensitivity to initialization and a lack of stable convergence. As a baseline, these results establish that a straightforward fully connected architecture operating on raw positional and elevation features can achieve moderate classification performance, setting the bar for the CNN and Bayesian models to improve upon.

A limitation of the MLP is its lack of interpretability. With over 21,000 individual weights, no straightforward mapping exists between input features and predicted class. A single-layer perceptron would permit direct inspection of the weight vector to understand each feature's contribution, but the depth required to achieve this level of accuracy does not allow transparency.

The CNN model achieves a training accuracy of 0.464 on 13,083 images and a test accuracy of 0.448 on 4,361 images, with 31,874 trainable parameters. It should be noted that due to an oversight in the data pipeline, the validation set used during training was identical to the test set, meaning the reported test accuracy does not reflect a truly held-out evaluation. The model was trained for relatively few epochs, as it is prone to heavy overfitting on the training data.

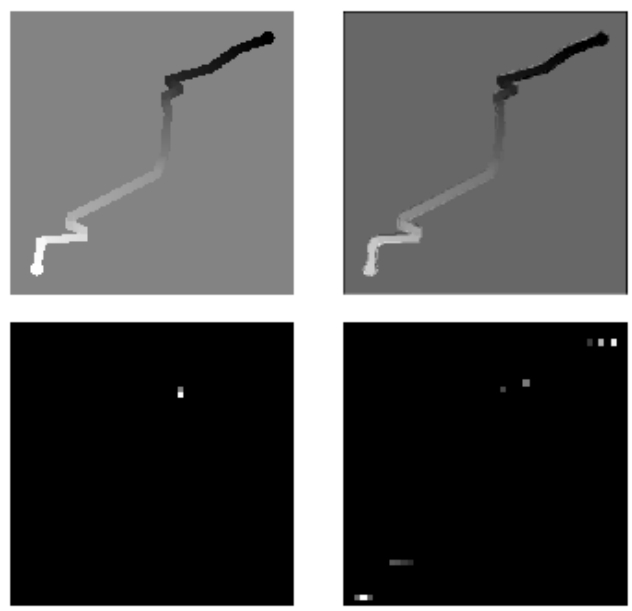

Fig. 3: Trail image passed through CNNs kernels

The four supplementary features (elevation gain, elevation loss, average grade, max grade) concatenated after the convolution layers improved test accuracy by two or three percentage points on average. While this gain is modest, the supplementary features also served to stabilize performance across a wider range of hyperparameter configurations, making the

model less sensitive to architectural tuning. These features are imputed directly from raw GPX data, preserving the model's applicability in interactive settings where a user uploads a file and receives a prediction.

That the CNN does not meaningfully outperform the MLP baseline, despite its substantially more complex architecture and image-based input representation, suggests that the rasterized trail images, while visually informative, may not encode difficulty-relevant signals beyond what is already captured beyond the raw positional and elevation features. A contribution to this is the absence of inertial measurement unit (IMU) data in the dataset. Langer et al. [9] demonstrated strong CNN performance for MTB trail classification using accelerometer and gyroscope stream, which capture surface roughness, vibration, and obstacle impacts, things not present in GPS recordings. Without this complementary signal, the additional representational capacity of the CNN has limited room to provide benefit over a simpler architecture.

Despite the CNN's modest accuracy gains, inspection of the learned convolutional filters reveals the model does extract meaningful spatial features from the trail images (3). The first convolutional layer applies a single 3x3 kernel that acts as a generalizing filter, smoothing the image and reducing the prominence of localized intensity peaks, such as the trail's starting point. Following a 5x5 max pooling, the second convolutional layer applies two kernels that learn distinct spatial patterns. The first kernel activates most strongly at sharp directional changes in the trail path, suggesting the model has learned that trail tortuosity is a relevant feature for difficulty classification. The second kernel responds to the endpoints of the trail as well as sharp turns, selectively highlighting the spatial extremes and directional changes in the original image. These activations, while not sufficient to drive substantial accuracy improvements over the baseline, confirm the CNN is learning difficulty-relevant spatial features rather than fitting to noise.

The Bayesian model achieves a training accuracy of 0.443 on 13,955 input vectors and a test accuracy of 0.445 on 3,489 input vectors. Using only four trail-level features, matching the CNN supplemental features (elevation gain, elevation loss, average grade, and max grade), this model performs comparably to both the MLP and CNN despite operating on a substantially reduced feature set. This result reinforces the finding of Steyn et al. [5] that gradient-related features are among the strongest predictors of trail difficulty.

The primary advantage of the Bayesian framework is the interopretability afforded by its prior weight distributions. (4) shows the prior distribution for the average grade weight across classes, using Difficult as the reference class. The Easy class exhibits a strong negative weight for average grade, consistent with the expectation that easy trails are characterized by low average gradients and thus sit furthest from the Difficult reference in this feature dimension. The Intermediate and Intermediate/Difficult classes show less extreme weight, reflecting their position between the two endpoints of the difficulty scale. The max grade feature follows a familiar

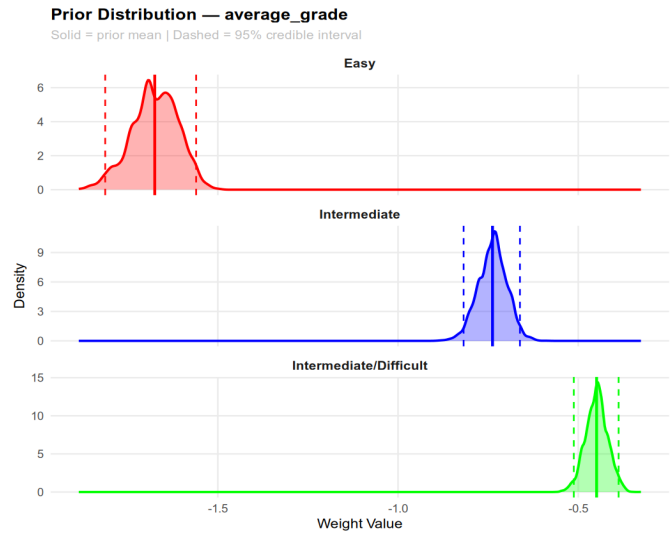

Fig. 4: Prior distribution for average grade weight

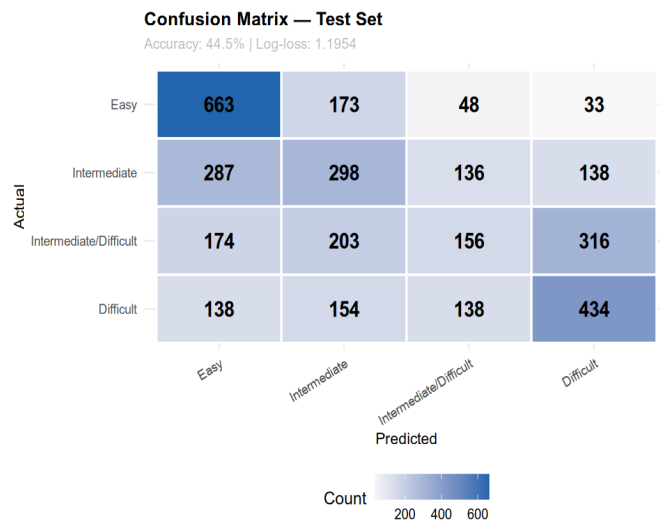

Fig. 5: Confusion matrix for the Bayesian model

pattern: the Easy class again shows the most negative prior weight relative to the reference class, with the remaining classes falling in ordinal order. The standard deviations of all prior weight distribution fall within the range of 0.024-0.070, indicating that the model has converged to relatively confident estimates for each feature's contribution.

It should be noted that the prior distributions were generated from only 6,150 retained samples due to the thinning and burn-in requirements described in Section III-C. Running the sampler for additional interactions or employing a more efficient algorithm would likely yield smoother prior and tighter credible intervals.

(5) presents the confusion matrix for the Bayesian model. The model shows a tendency to over-predict the Easy and Difficult classes relative to the intermediate categories, despite balanced training data (4,361 examples per class). This bias is

Model	Training Accuracy	Test Accuracy
MLP	0.534	0.500
CNN	0.464	0.448
Bayesian	0.435	0.445

TABLE I: Model Performance Comparison

partially attributed to the reference class architecture. With Difficult as the reference, the model learns the strongest discriminative signal for classes that are most distinct from the reference (i.e. Easy) while the intermediate classes, which share characteristics with both extremes, are more difficult to separate. This pattern is consistent with the weight distribution shown in (4), where the Intermediate and Intermediate/Difficult classes are compressed between Easy and Difficult in feature space.

All three models converge on test accuracies in the 0.44 - 0.50 range, exceeding the 0.25 baseline of random assignment across four classes but falling short of strong predictive performance. The convergence of three architecturally different models, in complexity and learning mechanisms, to similar accuracy levels suggests a ceiling imposed by the data itself rather than by any individual model's capacity. Trail difficulty ratings are inherently subjective, and the features extractable from GPX data, while correlated with difficulty, do not capture the full picture of physical trail characteristics that inform human ratings. This interpretation is consistent with the partial observability discussed in (II-B).

That the Bayesian model matches the performance of the more complex MLP and CNN using only four features is flabbergasting. It suggests that gradient and elevation statistics capture the majority of the difficulty-relevant signals available in GPX data, and that the spatial patterns learned by the CNN from rasterized trail images provide limited additional information beyond what these aggregate statistics already encode.

V. Discussion

This study investigated automated mountain bike trail difficulty classification using GPX data, comparing three modeling approaches: a multi-layer perceptron baseline, a convolutional neural network operating on rasterized trail images, and a Bayesian classifier with interpretable prior weight distributions. All three models achieved test accuracies in the 0.44-0.50 range across four consolidated difficulty classes, exceeding the 0.25 random baseline but falling well short of reliable predictive performance. The convergence of architecturally distinct models to similar accuracy levels points to a ceiling imposed by the data rather than by model capacity, consistent with the partial observability of difficulty-relevant trail features in GPX recordings.

The Bayesian model's comparable performance using only four gradient and elevation features suggests that these aggregate statistics capture the bulk of the difficulty-relevant signals available in GPX data. Its prior weight distributions further confirm that grade-related features are the strongest discriminators, with average and max grade showing the clearest separation between difficulty classes. The CNN's learned kernels demonstrate that spatial features such as trail tortuosity and directional changes are detectable through convolution on rasterized trail images, but these patterns do not translate into meaningful accuracy gains over simpler approaches. The MLP baseline, while achieving the highest test accuracy, exhibited substantial run-to-run variance, limiting its reliability as a standalone classifier.

Several methodological assumptions warrant discussion. The Bayesian model's Markov chain exhibited strong autocorrelation prior to thinning, partially violating the independence assumption underlying MCMC inference. While thinning and burn-in mitigated this, the resulting 6,150 retained samples from 1.5 million iterations represent a low effective sample size, and the posterior estimates would benefit from longer chains or more efficient samplers. For the CNN, the validation and test sets were inadvertently identical, meaning the reported test accuracy does not reflect evaluation on truly held-out data; this limits the strength of conclusions drawn from the CNN's generalization performance. The MLP's sensitivity to random initialization, producing a twofold difference in accuracy across consecutive runs, raises questions about the stability of the learned decision boundaries for this feature set and architecture.

The class consolidation and balancing strategy also introduces tradeoffs. Merging the original six difficulty classes into four reduced the classification resolution and may have grouped trails with meaningfully different characteristics, particularly at the boundaries between merged categories. Random subsampling to balance classes ensured an unbiased class prior during training but discarded a substantial portion of the majority-class data, potentially reducing the models' exposure to the full variability within those classes.

More broadly, the central limitation of this work is the data source itself. GPX files capture positional and elevation information but are blind to the physical trail characteristics that most directly determine technical difficulty: surface material, obstacle size and frequency, trail width, exposure, and constructed features such as drops and berms [1] [5] [6]. The subjective nature of the community-assigned difficulty labels introduces additional noise that no model architecture can fully resolve, as demonstrated by Siebert et al. 's [4] finding that riders of different skill levels perceive the same trail at different difficulty levels. These two factors together create a theoretical lower bound on achievable classification error for any GPX-based approach.

Several directions could improve upon these results in future work. Incorporating IMU data from handlebar or frame mounted sensors, as demonstrated by Langer et al. [9], would provide direct measurements of surface roughness, vibration intensity, and obstacle impacts, addressing the most significant gap in the GPX feature space. Additionally, augmenting GPX traces with remote sensing data, such as satellite imagery or LiDAR terrain models could provide proxy measurements for surface condition and trail width without requiring on-bike

instrumentation. An alternative approach could be focusing on personalized ratings, combining a rider's skill level, derived from historical riding traces, with these predictive models to rate trails on perceived difficulty, rather than predicting a single universal difficulty rating. This approach sidesteps the core limitation identified in this study, where a single difficulty rating cannot account for the wide variation in how riders of different skill levels perceive the same trail [4]. Instead of attempting to denoise subjective labels into one ground truth, a personalized system embraces the subjectivity by making the prediction relative.

Despite the modest classification accuracy, the models developed in this study demonstrate that GPX data alone contains sufficient signals to predict trail difficulty above chance, and that this prediction can be delivered though a web application requiring nothing more than a standard GPS recording. The Bayesian model in particular offers a framework that quantifies uncertainty in its predictions, well suited in a domain where subjectivity is inherent and overconfident ratings carry safety consequences.

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APPENDIX A
MLP IMPLEMENTATION

```
# pts is the user specified GPS observation points per trail

mlp = models.Sequential([
    layers.Input(shape=(pts, 3)),
    layers.Flatten(),
    layers.Dense(64, activation="relu"),
    layers.Dropout(0.2),
    layers.Dense(32, activation="relu"),
    layers.Dropout(0.2),
    layers.Dense(16, activation="relu"),
    layers.Dropout(0.2),
    layers.Dense(3, activation="softmax"),
])

mlp.compile(optimizer=Adam(),
            loss=CategoricalCrossentropy(),
            metrics=["accuracy"])
```

APPENDIX B
CNN IMPLEMENTATION

```
height, width, num_color_channels = img_size

input_image = Input(
    shape=(
        height,
        width,
        num_color_channels,
    )
)
extra_features = Input(shape=(num_extra_features,))

x = Conv2D(1, (3, 3), padding="same", activation="relu")(input_image)
x = MaxPooling2D((5, 5))(x)

x = Conv2D(2, (3, 3), padding="valid", activation="relu")(x)
x = MaxPooling2D((5, 5))(x)

post_convolution = Flatten()(x)
x = Concatenate()([post_convolution, extra_features])

x = Dense(128)(x)
x = LeakyReLU(alpha=0.1)(x)
x = Dropout(0.7)(x)

x = Dense(64)(x)
x = LeakyReLU(alpha=0.1)(x)
x = Dropout(0.7)(x)

x = Dense(32)(x)
x = LeakyReLU(alpha=0.1)(x)
x = Dropout(0.7)(x)

out_layer = Dense(num_classes, activation="softmax")(x)

model = Model(inputs=[input_image, extra_features], outputs=out_layer)
model.compile(
    loss="categorical_crossentropy", optimizer="adam", metrics=["accuracy"]
)
```

APPENDIX C
BAYESIAN MULTINOMIAL

```
log_likelihood <- function(W, X, y) {
  eta <- cbind(X %*% W, 0) # last col is reference class = 0
  row_max <- apply(eta, 1, max)
  log_sum <- log(rowSums(exp(eta - row_max))) + row_max
  sum(eta[cbind(seq_len(nrow(eta)), y)] - log_sum)
}
log_prior <- function(W) -0.5 * sum(W^2) / SIGMA_PRIOR^2
log_posterior <- function(W, X, y) log_likelihood(W, X, y) + log_prior(W)

run_mcmc <- function(X_train, y_train, feature_count, existing_weights_file) {
  W_dim <- feature_count * (class_count - 1)
  W_current <- matrix(0, feature_count, class_count - 1)

  current_log_probability <- log_posterior(W_current, X_train, y_train)

  n_keep <- floor((N_ITER - BURNIN) / THIN)

  W_samples <- array(NA, dim = c(n_keep, feature_count, class_count - 1))
  sample_idx <- 0
  n_accept <- 0

  for (iter in seq_len(N_ITER)) {
    epsilon <- matrix(rnorm(W_dim, 0, PROPOSAL_SD), feature_count, class_count - 1)
    W_prop <- W_current + epsilon

    log_probability <- log_posterior(W_prop, X_train, y_train)
    alpha <- log_probability - current_log_probability

    if (alpha > log(runif(1))) {
      W_current <- W_prop
      current_log_probability <- log_probability
      n_accept <- n_accept + 1
    }

    if (iter > BURNIN && ((iter - BURNIN) %% THIN == 0)) {
      sample_idx <- sample_idx + 1
      W_samples[sample_idx, , ] <- W_current
    }
  }

  acceptance_rate <- n_accept / N_ITER
  cat(sprintf("Final␣acceptance␣rate:␣%.2f%%\n", 100 * acceptance_rate))

  post_chain <- W_samples[seq_len(sample_idx), , , drop = FALSE]

  list(post_chain = post_chain,
    sample_idx = sample_idx,
    acceptance_rate = acceptance_rate)
}
```